Systematic Map Codebook — FT-R2a

ILRI / Bristlepine Resilience Consultants — 2025–2026

Where to find everything

→ Open the FT-R2a folder on Google Drive

What	Where
Papers to code	`FT-R2a pdfs/` subfolder inside the Drive folder above
Blank coding sheet	`coding_ft-r2a_template.csv` inside the Drive folder
Your completed sheet	Save as `coding_ft-r2a_INITIALS.csv` (e.g. `coding_ft-r2a_AZ.csv`) and upload back to the same folder

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How to code

1. Download `coding_ft-r2a_template.csv` and save a copy as `coding_ft-r2a_INITIALS.csv`
2. Open each paper from the `FT-R2a pdfs/` subfolder
3. Apply the five inclusion criteria (see below) — enter `confirmed_include` first
4. If `confirmed_include = yes`: fill in all 16 extraction fields for that row
5. If `confirmed_include = no`: leave extraction fields blank; note the reason in `notes`
6. Enter your initials in `coder_id` — same initials on every row
7. Upload your completed file back to the Drive folder when done

Rules:

- Code independently — do not discuss your decisions with anyone until instructed
- If a field is not reported in the paper: leave blank or enter `not_reported`
- If you are unsure: make your best judgement and add a note in the `notes` column

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Inclusion criteria

These papers have been retrieved but not yet formally screened. Read each paper and confirm it meets **all five criteria** before extracting data. The paper must satisfy ALL five — if it fails any one, mark `confirmed_include = no`.

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1. Population

Include if: Individuals or households engaged in small-scale agricultural production (crop, livestock, fisheries/aquaculture, agroforestry) in an LMIC. Also include marginalized producer groups (women, youth, landless farmers, indigenous peoples, ethnic minorities, migrant/seasonal workers).

Exclude if: Large-scale commercial agriculture, agribusiness without smallholder focus, or non-agricultural livelihoods only.

When in doubt: - Mixed samples (smallholders + larger farms): INCLUDE if smallholders are mentioned and analyzed - “Farmers” in LMIC contexts without explicit “smallholder” label: INCLUDE unless clearly commercial scale - Industrial or corporate agriculture: EXCLUDE

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2. Concept (Adaptation)

Include if: Study assesses or documents adaptation processes (learning, decision-making, adoption, participation, behavior change) OR adaptation outcomes (productivity, income, livelihoods, wellbeing, resilience, risk reduction).

Exclude if: - Climate impacts/vulnerability assessment ONLY (no adaptation action measured) - Mitigation only (reducing emissions) - General development or agronomy without explicit climate risk framing - Purely theoretical or conceptual (no intervention or measurement described)

When in doubt: - Adoption/uptake of climate-smart agriculture or water management: INCLUDE - Livelihood diversification in response to climate stress: INCLUDE - Vulnerability index or risk score without coping/adaptation responses: EXCLUDE - Farmer awareness/perception of climate change alone: EXCLUDE

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3. Context (Climate + Agriculture)

Include if: Agricultural setting (crop, livestock, fisheries, agroforestry) with explicit climate hazard link — drought, flooding, temperature extremes, rainfall variability, or other climate stressors affecting production.

Exclude if: - Non-agricultural sector - Agricultural study with no climate dimension - Climate mentioned only as background (not driving the research question) - Agricultural impacts/responses unrelated to climate

When in doubt: - “Water scarcity” or “aridity” in semi-arid regions (implies climate): INCLUDE - “Food insecurity” in LMIC rural contexts with adaptation responses: INCLUDE - Generic “livelihood change” in

non-climate context: EXCLUDE - Agriculture + environmental change (without climate specificity): EXCLUDE

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4. Methodology

Include if: Study describes, evaluates, or applies a measurable method or framework to assess adaptation (indicators, metrics, surveys, qualitative interviews, participatory assessments, models with empirical data). Published between 2005–2025.

Exclude if: - Purely theoretical/conceptual (no method or data) - Systematic reviews, meta-analyses, or literature reviews - Pre-2005 publication - Opinion pieces or policy briefs without empirical method

When in doubt: - Narrative case study of farmer adaptation: INCLUDE (qualitative empirical method) - Agent-based model with survey calibration: INCLUDE (model + data) - Pilot application of an adaptation tool: INCLUDE - Crop variety trial without farmer adoption analysis: EXCLUDE

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5. Geography

Include if: Study conducted in one or more Low- or Middle-Income Countries (LMICs) or Global South countries. Single or multi-country studies OK if at least one LMIC is involved.

Exclude if: OECD high-income countries only (USA, UK, EU, Australia, Canada, Japan, South Korea, etc.) with no LMIC component.

When in doubt: - China (all provinces): INCLUDE (upper-middle-income) - South Africa: INCLUDE (upper-middle-income) - Multi-country study (e.g. Peru + Germany): INCLUDE (Peru qualifies) - Brazil, Vietnam, Thailand, Iran: INCLUDE (all are LMIC)

Decision rule: Enter `yes` in `confirmed_include` if the paper meets ALL five criteria. Enter `no` if it fails any one criterion — leave extraction fields blank and briefly note which criterion failed in the `notes` column.

Quick reference

Field	Valid values
`confirmed_include`	`yes` `no`
`publication_type`	`journal_article` `report` `working_paper` `thesis` `other`
`geographic_scale`	`local` `sub-national` `national` `multi-country` `regional`
`producer_type`	`crop` `livestock` `fisheries_aquaculture` `agroforestry` `mixed` `undefined`
`marginalized_subpopulations`	`women` `youth` `people_with_disabilities` `landless` `indigenous_peoples` `ethnic_minorities` `migrant_seasonal_workers` `other` `none_reported`
`adaptation_focus`	**free text**
`process_outcome_domains`	`knowledge_awareness_learning` `decision_making_planning` `uptake_adoption` `behavioral_change` `participation_coproduction` `institutional_governance` `access_information_services` `yields_productivity` `income_assets` `livelihoods` `wellbeing` `risk_reduction` `resilience_adaptive_capacity`
`indicators_measured`	**free text**
`methodological_approach`	`qualitative` `quantitative` `participatory` `modeling_with_empirical_validation` `experimental`
`purpose_of_assessment`	`research` `project_learning` `program_evaluation` `donor_reporting` `national_reporting`
`data_sources`	`surveys` `administrative_data` `remote_sensing` `participatory_methods` `secondary_data` `other`
`temporal_coverage`	`cross_sectional` `seasonal` `longitudinal` `repeated_cross_sectional`
`cost_data_reported`	`yes` `no`
`strengths_and_limitations`	**free text**
`lessons_learned`	**free text**

Screening check

`confirmed_include`

Read the paper in full, then apply the five inclusion criteria from the cover page.

Value	Meaning
`yes`	Paper meets all five criteria — proceed to fill in fields 1–16
`no`	Paper fails one or more criteria — leave fields 1–16 blank; record reason in `notes`

The 16 coding fields

1. `publication_year`

Year the study was published. If multiple versions exist, use the most recent.

Valid values: integer — e.g. `2021`

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2. `publication_type`

Value	Meaning
`journal_article`	Peer-reviewed journal
`report`	Project/programme report, evaluation report, policy brief
`working_paper`	Pre-publication working paper or discussion paper
`thesis`	PhD or master's dissertation
`other`	Book chapter, conference paper, dataset paper

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3. `country_region`

Country or countries where the study was conducted. Use a region label only if no countries are named.

Format: free text; separate multiple entries with semicolons

Examples: `Tanzania` | `Ghana; Mali; Burkina Faso` | `East Africa` | `Global`

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4. `geographic_scale`

Value	Meaning
`local`	Village, community, single site
`sub-national`	District, province, region within a country
`national`	One country
`multi-country`	Two or more specific named countries
`regional`	Broad region without country-level focus

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5. `producer_type`

Select all that apply; separate with semicolons.

Value	Meaning
`crop`	Crop farming systems
`livestock`	Livestock farming systems
`fisheries_aquaculture`	Fisheries or aquaculture systems
`agroforestry`	Agroforestry systems
`mixed`	Explicitly mixed crop-livestock
`undefined`	Generic “smallholder farmers” with no production system specified

Where to look: Abstract and Introduction (authors state target population and agricultural focus early); Methods / Study Area / Site Description (most reliable — authors describe the agricultural baseline of communities surveyed).

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6. `marginalized_subpopulations`

Code only what is **explicitly stated** in the paper — do not infer. Select all that apply; separate with semicolons.

Value	Meaning
`women`	Women explicitly targeted or gender-disaggregated analysis
`youth`	Young farmers or age-disaggregated outcomes
`people_with_disabilities`	Persons with disabilities explicitly named
`landless`	Landless households or tenure-insecure groups

Value	Meaning
`indigenous_peoples`	Indigenous communities named
`ethnic_minorities`	Ethnic minority groups named
`migrant_seasonal_workers`	Migrant or seasonal agricultural workers
`other`	Any other marginalised group explicitly named — add a note in `notes`
`none_reported`	No marginalised groups mentioned

Where to look: Abstract and Introduction (often stated immediately if the study targets a vulnerable subgroup); Methods / Sampling (look for purposive sampling criteria, e.g. “deliberately selected based on gender or youth status”).

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7. `adaptation_focus` *(free text)*

The specific climate adaptation action, intervention, or practice the study tracks. Write 5–15 words based on what the paper describes.

Examples: - `crop varieties and genetics` - `water management and irrigation` - `climate information and advisory services` - `livelihood diversification (non-farm)` - `agroforestry` - `livestock management` - `financial strategies` - `social and community networking`

Where to look: Abstract and Introduction (primary interventions highlighted early); Methods / Study Design (specific strategies tested or surveyed); Results tables (adoption rates and regression variables list exact practices tracked); Discussion and Conclusion (if the study is a modeling or vulnerability assessment, the adaptation focus may be proposed here rather than tested); Appendix / Survey Instrument (if included, shows exact adaptation options asked about).

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8. `process_outcome_domains`

What the study measures. Select all that apply; separate with semicolons.

Rule: Code only domains that are **explicitly measured** — not domains that could be inferred. If the paper measures yield but not income, code `yields_productivity` only.

Value	Meaning
`knowledge_awareness_learning`	Awareness of climate change, knowledge of options, learning outcomes
`decision_making_planning`	Farmer decision-making, use of forecasts in planning
`uptake_adoption`	Rates of adoption of a practice or technology
`behavioral_change`	Changes in farming or land management behaviour

Value	Meaning
`participation_coproduction`	Participation in programmes, co-design, farmer group engagement
`institutional_governance`	Policy uptake, institutional change, governance
`access_information_services`	Access to extension, credit, markets, climate information
`yields_productivity`	Crop yield, livestock productivity, fish catch
`income_assets`	Household income, asset accumulation
`livelihoods`	Food security, livelihood diversification
`wellbeing`	Health, nutrition, subjective wellbeing
`risk_reduction`	Reduced exposure or sensitivity to climate shocks
`resilience_adaptive_capacity`	Adaptive capacity indices, resilience scores, vulnerability indices

Where to look: Methods / Variable Descriptions / Summary Statistics tables (specific indicators listed); Results (qualitative and quantitative changes reported here); Discussion and Conclusion (how the intervention affected social processes, learning, or institutional dynamics).

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9. `indicators_measured` *(free text)*

The specific indicators or metrics used. Extract from the methods or results sections. Include units where stated.

Examples: - `yield (t/ha), income (USD/season), food security score (HFIAS)` - `adoption rate (%), area under improved variety (ha)` - `climate resilience index score`

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10. `methodological_approach`

Select all that apply; separate with semicolons.

Value	Meaning
`qualitative`	Qualitative research design (interviews, focus groups, ethnography)
`quantitative`	Quantitative research design (surveys, statistical analysis)
`participatory`	Only when participation is the **primary** design (PRA, photovoice) — not when farmers are merely surveyed
`modeling_with_empirical_validation`	Crop/climate/agent-based models validated with field data

Value	Meaning
`experimental`	RCTs, natural experiments, quasi-experimental approaches

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11. `purpose_of_assessment`

Value	Meaning
`research`	Academic study with no programme affiliation
`project_learning`	M&E embedded in an ongoing project
`program_evaluation`	Formal evaluation of a completed programme
`donor_reporting`	Report produced for a donor or funder
`national_reporting`	Contributing to NAPs, NDCs, or national monitoring frameworks

Where to look: Abstract (high-level summary of why the study was conducted); Introduction (research questions or objectives are typically stated explicitly at the end of the introduction); Conclusion (authors restate core purpose when summarising findings).

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12. `data_sources`

Select all that apply; separate with semicolons.

Value	Meaning
`surveys`	Primary household or farm surveys
`administrative_data`	Government or programme records
`remote_sensing`	Satellite or aerial imagery
`participatory_methods`	PRA, focus groups, photovoice as a data source
`secondary_data`	Existing datasets, literature, or other secondary sources
`other`	Any data source not listed above

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13. `temporal_coverage`

Value	Meaning
`cross_sectional`	Single point in time or single season

Value	Meaning
`seasonal`	Intra-annual, across one or a few seasons
`longitudinal`	Multi-year, panel data, repeated measures with the same individuals
`repeated_cross_sectional`	Multi-year, repeated measures with different individuals each time

Where to look: Methods / Data Collection / Study Design (most definitive — authors detail when and how often data were collected); Data and Variables / Climate Modeling sections (for studies using secondary data or historical records, these define the time period analysed).

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14. `cost_data_reported`

Value	Meaning
`yes`	Paper explicitly reports cost figures or cost-effectiveness ratios
`no`	No cost data — a qualitative mention of “limited resources” counts as `no`

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15. `strengths_and_limitations` *(free text)*

Author-reported strengths and limitations. Look in the Discussion, Conclusion, and Limitations sections. Extract verbatim where possible.

Format: `Strength: <text>. Limitation: <text>.`

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16. `lessons_learned` *(free text)*

Key lessons or recommendations the authors report. 1–3 sentences. Focus on what they say should be done differently — not the findings themselves. Enter `not_reported` if absent.

Where to look: Discussion (primary area where authors interpret results and extract practical insights); Conclusion / Implications / Recommendations subsections; Abstract (final sentences often distill the most critical lesson).

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